



Service Data

SD-29-50022

Bendix® FLR-25™ Radar Sensor

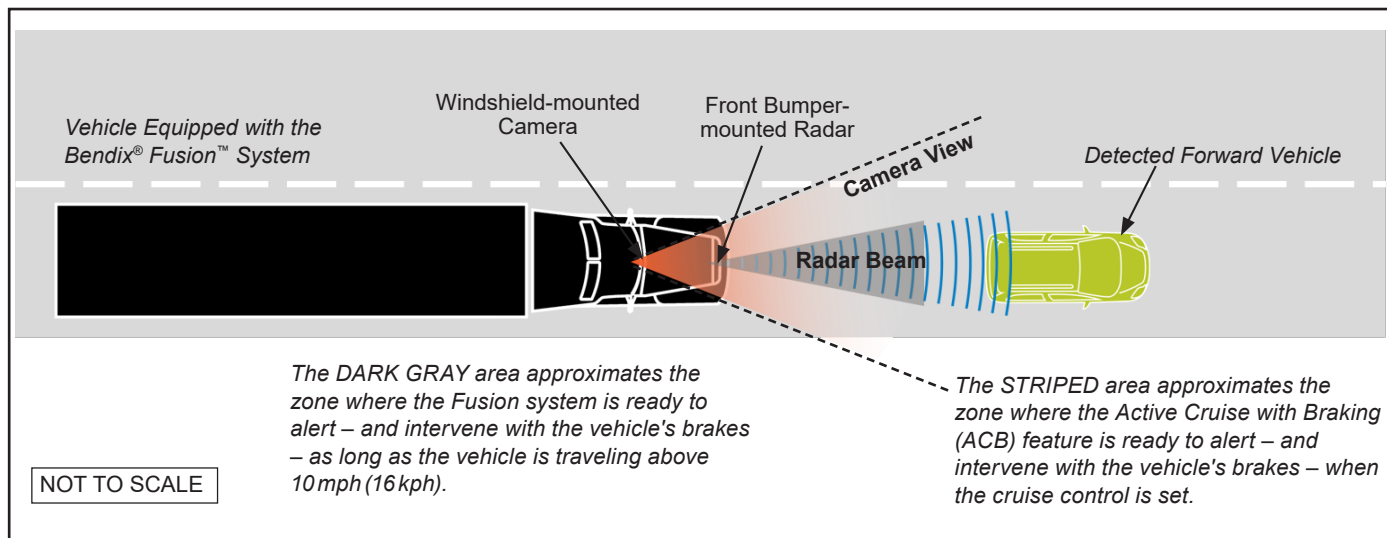


Figure 1 – Bendix® Fusion™ System

! IMPORTANT

Bendix®-brand Electronic Control Units (ECUs) are not designed to store data for purposes of accident reconstruction, and Bendix® ACom® Diagnostic Software is not intended to retrieve data for purposes of accident reconstruction. Bendix makes no representations as to the accuracy of data or video retrieved and interpreted from ECUs for purposes of accident reconstruction. Bendix does not offer accident reconstruction services or interpretation of stored data. Bendix ECUs are not protected from fire, loss of power, impact damage, or other conditions that may be sustained in a crash situation and may cause data to be unavailable or irretrievable.

! IMPORTANT

The Bendix Fusion system will not operate and will set a Diagnostic Trouble Code (DTC) if any of the following vehicle systems also show a relevant DTC: engine, engine cruise, instrument cluster, Bendix® ABS, Bendix® ATC, Bendix® ESP®, or transmission. These components must be repaired and cleared of DTCs before troubleshooting the Fusion system. NOTE: Clearing the vehicle DTCs may be the only step needed to reestablish full Fusion system functionality.

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To search DTCs and receive additional troubleshooting support, refer to:
B2Bendix.com > Services & Support > Diagnostic Trouble Code Search

GENERAL SAFETY GUIDELINES



WARNING! PLEASE READ AND FOLLOW THESE INSTRUCTIONS



TO AVOID PERSONAL INJURY OR DEATH:

When working on or around a vehicle, the following guidelines should be observed AT ALL TIMES:

- ▲ Park the vehicle on a level surface, apply the parking brakes, and always block the wheels. Always wear personal protection equipment.
- ▲ Stop the engine and remove the ignition key when working under or around the vehicle. When working in the engine compartment, the engine should be shut off and the ignition key should be removed. Where circumstances require that the engine be in operation, **EXTREME CAUTION** should be used to prevent personal injury resulting from contact with moving, rotating, leaking, heated, or electrically charged components.
- ▲ Do not attempt to install, remove, disassemble, or assemble a component until you have read, and thoroughly understand, the recommended procedures. Use only the proper tools and observe all precautions pertaining to use of those tools.
- ▲ If the work is being performed on the vehicle's air brake system, or any auxiliary pressurized air systems, make certain to drain the air pressure from all reservoirs before beginning ANY work on the vehicle. If the vehicle is equipped with a Bendix® AD-IS® air dryer system, a Bendix® DRM™ dryer reservoir module, a Bendix® AD-9si®, AD-HF®, or AD-HFi® air dryer, be sure to drain the purge reservoir.
- ▲ Following the vehicle manufacturer's recommended procedures, deactivate the electrical system in a manner that safely removes all electrical power from the vehicle.
- ▲ Never exceed manufacturer's recommended pressures.
- ▲ You should consult the vehicle manufacturer's operating and service manuals, and any related literature, in conjunction with the Guidelines above.
- ▲ Never connect or disconnect a hose or line containing pressure; it may whip and/or cause hazardous airborne dust and dirt particles. Wear eye protection. Slowly open connections with care, and verify that no pressure is present. Never remove a component or plug unless you are certain all system pressure has been depleted.
- ▲ Use only genuine Bendix® brand replacement parts, components, and kits. Replacement hardware, tubing, hose, fittings, wiring, etc. must be of equivalent size, type, and strength as original equipment and be designed specifically for such applications and systems.
- ▲ Components with stripped threads or damaged parts should be replaced rather than repaired. Do not attempt repairs requiring machining or welding unless specifically stated and approved by the vehicle and component manufacturer.
- ▲ Prior to returning the vehicle to service, make certain all components and systems are restored to their proper operating condition.
- ▲ For vehicles with Automatic Traction Control (ATC), the ATC function must be disabled (ATC indicator lamp should be ON) prior to performing any vehicle maintenance where one or more wheels on a drive axle are lifted off the ground and moving.
- ▲ The power **MUST** be temporarily disconnected from the radar sensor whenever any tests **USING A DYNAMOMETER** are conducted on a vehicle equipped with a Bendix® Wingman® system.



WARNING

Improper use of the Bendix® Fusion™ Driver Assistance System can result in a collision causing property damage, serious injuries, and/or death.



WARNING

Due to the inherent limitations of image recognition technology, camera-based safety technology — on rare occasions — may not be able to detect or may misinterpret lane markings. At these times, alerts may not occur, or erroneous alerts may occur.



WARNING

Bendix safety technologies complement safe driving practices. No commercial vehicle safety technology replaces a skilled, alert driver exercising safe driving techniques and proactive, comprehensive driver training. Never wait for the system to intervene. Responsibility for the safe operation of the vehicle remains with the driver at all times.



IMPORTANT

It is the responsibility of the driver to remain vigilant and change driving practices depending on traffic and road conditions.

The Bendix™ FLR-25™ radar sensor is an integral component of the Bendix® Fusion™ Driver Assistance System, which integrates a camera, radar, and brakes. The Fusion system provides the following alerts and actions:

- Autonomous Emergency Braking (AEB)
 - Stationary Vehicle Braking (SVB)
 - Multi-lane AEB
 - Close-range AEB
- Highway Departure Braking (HDB)
- Active Cruise with Braking (ACB)
 - ACB Pedestrian Abort
- Overspeed Alert and Action (OAA)
- Alert Prioritization

NOTE: Depending on configuration, the system may record data and video of certain events and, when integrated with a telematics provider, can transmit this information for viewing on the SafetyDirect® by Rand McNally® Web Portal Processor. For support with SafetyDirect, contact Rand McNally at safetydirectsupport@randmcnally.com.

The major components that are used in the Bendix Fusion Driver Assistance System are:

- Bendix FLR-25 Radar Sensor
- Bendix® ESP® EC-80™ Controller
- SafetyDirect® Processor
- Bendix™ FLC-20™ or FLC-25™ Camera

For additional information, *refer to the following related documents on B2Bendix.com:*

- SD-13-4986, Bendix™ ESP® EC-80™ Controller
- SD-65-21025, SafetyDirect® Processor by Rand McNally®
- SD-64-20124, Bendix™ FLC-20™ Camera
- SD-29-50024, Bendix™ FLC-25™ Camera
- BW8107, Bendix® Safety Systems Challenging Scenarios Overview
- BW5266, Bendix® Fusion™ 3: Radar (FLR-25™) and Camera (FLC-25™) Calibration and Alignment

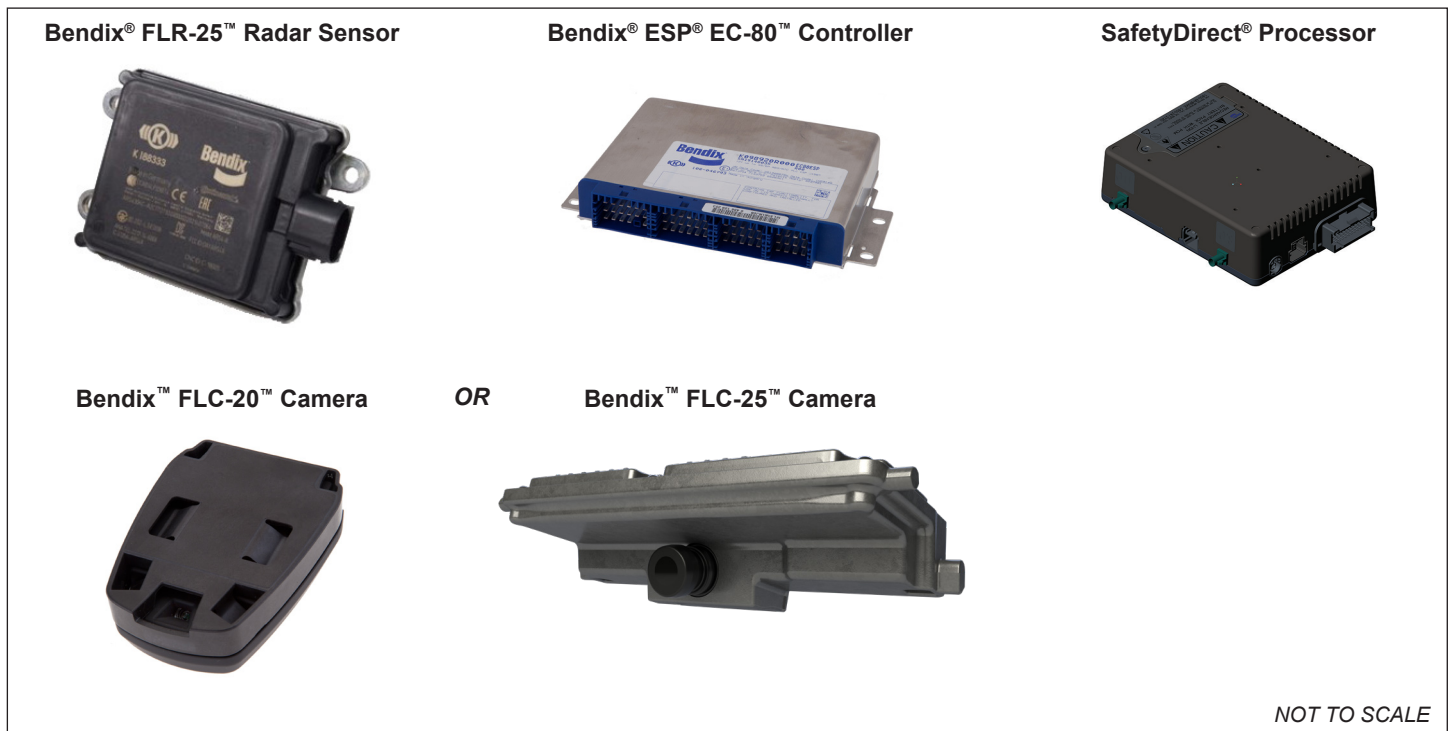


Figure 2 – Major Fusion System Components

1.0 INTRODUCTION TO TROUBLESHOOTING

Read the entirety of this document before performing any troubleshooting.

Diagnostic information is available using Bendix® ACom® Diagnostic Software. ACom Diagnostic Software is a PC-based program that is designed to meet RP1210 industry standards developed by the Technology and Maintenance Council (TMC). This software provides the technician with access to all available Fusion system diagnostic information and configuration capability, including:

- Electronic Control Unit (ECU) information
- Diagnostic Trouble Codes (DTCs) and repair information
- DTC reports

To search DTCs and receive additional troubleshooting support, refer to *B2Bendix.com* > *Services & Support* > *Diagnostic Trouble Code Search*.

For additional information on ACom Diagnostic Software and to purchase the software, *refer to B2Bendix.com*.

NOTE: Prior to troubleshooting, all aftermarket-installed components should be removed from the vehicle and the vehicle must be returned to factory specifications. Once the vehicle is returned to factory specifications, the fault should be recreated prior to performing troubleshooting steps.

1.1 FUSION SYSTEM TROUBLESHOOTING BASICS

Refer to Table 1 and the applicable Fusion Operator's Manual on B2Bendix.com.

Troubleshooting Basics	
Questions/Instructions	Next Steps
1. Does the driver hear a long warning beep?	If no problem is found and the test is passed, no additional beeps/lamps will be displayed nor will a DTC be set. If the system has found an issue that will prevent it from functioning properly, a long warning beep will sound to alert the driver, and a DTC will be logged in the system (typically with a status indicator/dash icon illuminated). For descriptions of all DTCs, <i>refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search</i> .
2. Have the driver describe the system behavior that they believe shows it is not working properly.	When diagnosing the system, especially in cases where there are no DTCs logged, find out which part of the system behavior appears to be operating improperly. NOTE: False brake applications are often misdiagnosed and are, in fact, <i>not</i> brake applications, but rather loss of engine throttle caused by a J1939 communication failure on the vehicle. This is frequently caused by aftermarket components installed in the vehicle CAN line (i.e. Electronic Drivers Logs [EDLs], GPS, or telematics systems).
3. Fill out the Troubleshooting Checklist on page 6 and generate a DTC report.	The Checklist and DTC report will clarify the problem and will be necessary if a call to the Bendix Tech Team is needed. <i>Refer to Section 1.2: Narrowing Down the Problem.</i>
Call the Bendix Tech Team at 1-800-AIR-BRAKE (1-800-247-2725), option 2, for troubleshooting assistance.	

Table 1 – Troubleshooting Basics

Troubleshooting Checklist

WARNING: The Troubleshooting Checklist should be completed by a qualified technician and should not be completed by the driver while the vehicle is in motion.

Refer to Section 1.2, Narrowing Down the Problem.

Detailed Scenarios and Tests	Record Driver's Answers for Follow-up with Bendix
Are there any active faults or lamps illuminated on the dash?	Yes ☐ No ☐ _____
Are there any aftermarket components installed on the vehicle?	Yes ☐ No ☐ _____
Does the vehicle maintain its set speed when the cruise control is switched on and set?	Yes ☐ No ☐ _____
Is the cruise control "set" icon displayed?	Yes ☐ No ☐ _____
While following a forward vehicle within radar range and the cruise control is switched on and set, record your response to the following: Is the "detected forward vehicle" icon displayed? What color is the icon? When the forward vehicle slows down, does the truck also slow down to maintain the set distance?	Yes ☐ No ☐ _____ Icon Color _____ Yes ☐ No ☐ _____
With engine cruise "off" and a forward vehicle present, does the audible alert become faster as the truck moves closer to the forward vehicle?	Yes ☐ No ☐ _____
With cruise control switched on and set, when the forward vehicle slows moderately or cuts in front of the truck and slows, did the driver observe any of the following conditions? Is there any indication of a DTC (example: dash lamp remains illuminated; DTC logged in Bendix® ACom® Diagnostic Software)? Does the vehicle slow and the Fusion system maintain the following distance? Is the engine throttle reduced? Is the engine retarder applied? Are the foundation brakes applied? Does the truck proceed toward the forward vehicle without a following distance alert or braking intervention?	Yes ☐ No ☐ _____ Yes ☐ No ☐ _____ Yes ☐ No ☐ _____ Yes ☐ No ☐ _____ Yes ☐ No ☐ _____ Yes ☐ No ☐ _____
With cruise engaged, when your vehicle passes a slower vehicle on the left or right on a straight road: Does the Fusion system ignore the vehicle you are overtaking? Does it give a following distance alert?	Yes ☐ No ☐ _____ Yes ☐ No ☐ _____
With cruise engaged, if a faster vehicle passes your vehicle on the left or right on a straight road: Does it give a Following Distance Alert (FDA)?	Yes ☐ No ☐ _____

1.2 NARROWING DOWN THE PROBLEM

Use Table 2 to help assess if the Fusion system is not performing correctly. Be sure to have a thorough understanding of the system's normal behavior; this will reduce the troubleshooting time. This table provides a guide to basic troubleshooting questions and possible corrective actions. Items *in italics* cross-reference to the service procedures in this manual to repair the condition described.

Narrowing Down the Problem	
Questions	Next Steps
Blocked Radar Sensor Issues	
Is mud, ice, or snow covering the radar sensor? Is anything blocking the view of the radar sensor?	Clean the radar sensor front surface immediately. Carefully remove anything blocking the radar sensor then power cycle and read any remaining DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i> <i>Refer to Appendix A3: FLR-25 Radar Sensor Mounting Clearance.</i> If the radar sensor is blocked by ice, snow, mud, tampering, etc. so that it cannot "see" a forward vehicle, the Fusion system will notify the driver by an audible and/or visual alert. After the blockage is removed, the audible and/or visual alert will be removed. Drivers should visually check the radar sensor for blockage during the pre-trip inspection.
Potential False Warnings	
Do false warnings seem to happen in construction zones?	Several road scenarios have a tendency to cause false warnings, including construction zones and bridges. Unless these false warnings are frequent, the system is likely reacting normally. The driver should not set the cruise control in construction zones. If driver complaints persist, continue asking questions to more narrowly define the driving condition presenting the problems. Review proper operating conditions in the operator's manual.
Mounting Problems	
Is the radar sensor mounting location (bumper or cross-member) damaged? Does the system seem to not "see" as far as it "used to," or warn on many more overhead bridges/signs than previously?	Inspect the radar mounting. A solid mounting surface is necessary in order to hold the alignment. If the bumper or mounting cross-member is damaged, replace it. If either or both of these are replaced, care should be exercised to make sure the original radar mounting location is not changed, as that may cause alignment or performance issues.
Does the mounting bracket look damaged or tampered with?	Other than expected surface scratches or some discoloration over time, there should be no visible damage to the radar sensor bracket assembly or the bumper. The radar sensor needs a solid mounting surface in order to hold the alignment. If the radar sensor alignment cannot be held in place, the bracket assembly must be replaced, and/or if the bumper or mounting cross-member is damaged, replace it as needed.

Table 2 – Narrowing Down the Problem

Narrowing Down the Problem	
Questions	Next Steps
Has the system worked without problems in the past but is not working as expected now?	This is a good indication that something has changed. Review the following questions with the driver to further diagnose the problem.
Has the radar sensor been changed recently?	If so, the new radar sensor may be incompatible with the vehicle. In addition, check any system DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i> Check the radar sensor baud rate. Confirm the part number (located on the front face of the radar sensor) is the correct part number per the vehicle's VIN.
Did the radar sensor currently on the vehicle come from another vehicle?	The radar sensor may be incompatible with the new vehicle. Check system DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i>
With cruise control set, does the system consistently apply the foundation brakes when a forward vehicle slows?	This is normal operation. If the radar system interventions are not typical, the radar sensor may be misaligned. Inspect the radar mounting. A solid mounting surface is necessary in order to hold the alignment. If the bumper or mounting cross-member is damaged, replace it. The service technician will need to check DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i>
Does a DTC seem to occur when driving through the desert or in barren areas (no road signs, trees, or vehicles)?	In normal operation, the ACB feature of the Fusion system may indicate a DTC if it hasn't detected a metallic object after a pre-determined period. This is rare, but most likely to occur when driving in deserts or barren areas. If the system does set a DTC, the Fusion system provides a visible warning to the driver, but cruise control will still be active. No further troubleshooting is required. Full and proper radar sensor functionality will be restored after a power cycle.
Does the system seem to disengage after an automatic braking event?	This is normal operation. Contact the vehicle OEM for details on cruise control operation.
Does cruise control disengage sometimes when the brakes come on and not at other times?	This is normal operation. When traveling with lightly loaded trailers, or "bobtail", the ACB feature of the Fusion system may continue to function even after an automatic brake application. No driver input is needed. Contact the vehicle OEM for details on cruise control operation.
Does the connector or wiring appear damaged?	Wires can become corroded if the radar sensor is not plugged in properly. If the wires are corroded or damaged, replace the wire harness and clean the radar sensor. Check system DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i> <i>Refer to Section 2.5: Troubleshooting Wiring Harnesses.</i>
Does the system generate a DTC going down a grade when using ACB to slow the vehicle, but the DTC goes away later?	This is normal operation. The ACB feature of the Fusion system is not intended to be used on grades. Verify there are no DTCs using ACom Diagnostic Software. Proper downgrade driving techniques should be used. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i>
Does the radar sensor have noticeable damage beyond normal discoloration or surface scratches?	The radar sensor and bracket are very durable. However, if the radar sensor housing or cover is cracked or broken, immediately replace the damaged radar sensor and confirm there are no DTCs using ACom Diagnostic Software. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i> <i>Refer to Appendix A3: Radar Sensor Mounting Clearance.</i>

Table 2 – Narrowing Down the Problem

1.3 OVERVIEW OF POSSIBLE ISSUES

Some customer issues are actually misunderstandings of how the Fusion system performs normally. *Use Table 3 and refer to the applicable operator's manual on B2Bendix.com* to learn the causes of potential issues if the Fusion system is not performing correctly. Some issues can be investigated by a visual inspection. Others may cause a DTC to be logged. *Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.*

Overview of Possible Issues	
Issue	Description
Vehicle DTCs	The Fusion system will not operate and will set a DTC if any of the following vehicle systems also show a relevant DTC: engine, engine cruise, instrument cluster, ABS, ATC, ESP®, or transmission. These components must be repaired and cleared of DTCs before troubleshooting the Fusion system. NOTE: Clearing the vehicle DTCs may be the only step needed to reestablish full Fusion system functionality. For ABS, ATC, ESP, or other Bendix-related DTCs, <i>refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i> All other vehicle components should be researched using the OEM's defined practices.
System familiarity	Verify the system functionality. Is it operating normally or not? Drivers who are unfamiliar with the system may report dissatisfaction over the way it beeps or how it activates the brakes. <i>Refer to Section 1.0: Introduction to Troubleshooting, Section 1.1: Fusion System Troubleshooting Basics, and the Diagnostic Trouble Code Search on B2Bendix.com – if necessary – to verify if the system is functioning normally;</i> then continue. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i>
DTCs caused by temporary operating conditions	Some DTCs indicate a temporary condition and will clear when that condition is no longer present. If these persist, further investigation is warranted. <i>Refer to Section 1.1: Fusion System Troubleshooting Basics.</i>
Radar sensor blocked	If the system does not seem to work at all and the dash display does not show a forward vehicle, the radar sensor may possibly be blocked. If the radar is blocked, an alert should be issued to the driver to indicate this condition. Visually inspect the area around the radar, clear the blockage, turn the ignition on, and run through a power cycle. <i>Refer to Appendix A3 for more information about radar mounting and clearances.</i>
Damaged radar sensor or bracket	If the vehicle has been in an accident, it is likely the radar sensor will need to be re-inspected or replaced. Inspect the radar sensor and housing for damage. Radar sensor discoloration or small scratches may be acceptable. Significant damage (such as cracks or broken pieces) will require radar sensor replacement. Regardless of the exterior condition, check for DTCs using ACom Diagnostic Software and the Diagnostic Trouble Code Search on B2Bendix.com to determine if radar sensor replacement is necessary. <i>Refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.</i>
Damaged connector or wiring	Visually inspect the connector and wire harness for corrosion or chafing. <i>Refer to Sections 2.1: Troubleshooting DTCs: Power Supply and 2.2 Serial Data (J1939) Communications Link of this document for additional troubleshooting.</i>
Radar sensor misalignment	Inspect the front of the vehicle. If a) it has been damaged; or b) the vehicle does not track straight, either of these conditions must be repaired before troubleshooting the Fusion system. Additionally, check the mounting brackets and vehicle cross members for misalignment or damage. If there is a DTC set or if the system does not function, the radar sensor may be severely misaligned and the Fusion system will not operate until this is corrected.
J1939 network problems	If the entire system is non-functional or intermittent J1939 errors occur, first check ACom Diagnostic Software to confirm that all Fusion components are being populated. If needed, <i>follow the instructions in Section 2.2: Serial Data (J1939) Communications Link.</i>
Power to radar sensor problems	If the entire system is non-functional, another likely cause may be a lack of power to the radar sensor. First check ACom Diagnostic Software to confirm that all Fusion components are being populated. If needed, <i>follow the instructions in Section 2.1: Troubleshooting DTCs: Power Supply.</i>

Table 3 – Overview of Possible Issues

2.0 TROUBLESHOOTING AND DIAGNOSTICS



All vehicle DTCs related to the engine, transmission, instrument cluster, engine cruise control, and Bendix ABS, ATC, or ESP systems must first be resolved—with no DTCs present during the vehicle operation while in cruise control—before attempting to diagnose Fusion system DTCs.

Important examples are VDC2 and EBC DTCs which are typically related to the brake controller. The hierarchy of ECUs determines that any DTCs on the brake controller must be resolved before attempting to troubleshoot the Fusion system.

2.1 TROUBLESHOOTING DTCS: POWER SUPPLY

IGNITION VOLTAGE TOO LOW

Measure the ignition voltage under load. Ensure that the ignition voltage is greater than 10 VDC (Volts DC). Check the vehicle battery and associated components. Inspect for damaged wiring, damaged or corroded connectors, and loose connections. Check the condition of the fuse.

IGNITION VOLTAGE TOO HIGH

Measure the ignition voltage. Ensure that ignition voltage is not greater than 16 VDC. Check the vehicle battery and associated components. Inspect for damaged wiring, damaged or corroded connectors, and loose connections.

POWER SUPPLY TESTS

1. Take all measurements at the radar sensor harness connector.
2. Place a load (i.e. 1157 stop lamp) across the supply voltage and ground connection. Measure the voltage with the load. The voltage on pin 1 to ground pin 8 should measure between 10 to 16 VDC.
3. Check for damaged wiring, damaged or corroded connectors, and loose connections.
4. Check the condition of the vehicle battery and associated components. Ensure the connection to ground is secure and tight.
5. Using the procedures described by the vehicle manufacturer, check the alternator output for excessive noise.

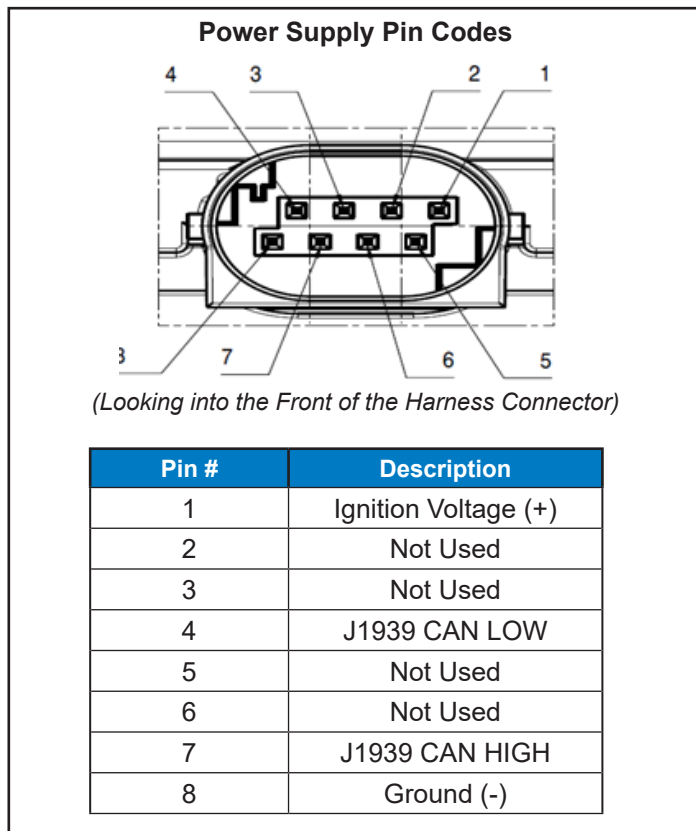


Table 4 – Harness Connector Pins

2.2 SERIAL DATA (J1939) COMMUNICATIONS LINK

Check for a loss of communications between the FLR-25 radar sensor, the ABS controller, the engine ECU, and other devices connected to the J1939 link. Check for damaged or reversed J1939 wiring. Check for corroded or damaged connectors and loose connections. Using procedures described by the vehicle manufacturer, verify the presence of the engine ECU and the ABS controller on the J1939 link. This can be done in ACom Diagnostic Software.

Verify the engine ECU configuration. Check for other devices inhibiting J1939 communications.

NOTE: The FLR-25 radar sensor will not report newly active J1939 DTCs until the engine has been running for 15 seconds. Do not attempt to diagnose J1939 DTCs with the engine running.

2.3 ENGINE COMMUNICATIONS (J1939) TEST PROCEDURE

NOTE: Remove all non-factory-equipped components – such as EDLs, GPS, and telematic systems – prior to testing communications. Clear DTCs. If the DTC does not return, the issue was likely caused by the removed component.

The FLR-25 radar sensor requires several J1939 messages from the engine ECU to control the engine and retarder torque for distance control and braking. The Fusion system will set a DTC if one of these messages is not present.

Use the engine manufacturer's diagnostic test procedures to verify that there are no errors present in the engine that may prevent the Fusion system from controlling the engine or retarder torque.

2.4 J1939 TROUBLESHOOTING PROCEDURE

1. Take all measurements at the harness connector unless otherwise indicated.



Do not insert probes into the back of the connector as this will damage the seal around the wire.



Do not insert any probe into the pin on the mating connector of the radar sensor that is greater than the dimension of the mating connector. Damaged connector pins will require the replacement of the harness.

2. Check for damaged or reversed J1939 wiring.

If the J1939 HIGH or J1939 LOW wiring circuits are damaged, such as shorting together, the entire J1939 link will be lost. The problem may be intermittent, enabling the J1939 link to sometimes operate normally. If this occurs, multiple DTCs will be logged in multiple engine and vehicle controllers.

If the J1939 HIGH and J1939 LOW wiring circuits are reversed, communication over the entire J1939 link will not be lost. Only those devices that are outside of the problem point from other devices will not receive, or be able to transmit, data messages.

3. Check for corroded or damaged wiring connector problems such as opens or shorts to voltage or ground. If the connector terminals are corroded, this may be an indication of water intrusion into the wiring system and possibly into the radar sensor. Replacement of the entire harness is recommended. If the terminals of the radar sensor are corroded, replacement of the radar sensor is recommended.
4. Check for other J1939 devices which may be inhibiting J1939 communication. The service technician should consult the vehicle manufacturer's procedures for other J1939 troubleshooting procedures. The device's power should be removed and measurements made at the ECU pins for shorts to ground and power pins and resistance between the J1939 HIGH or J1939 LOW input circuits.
5. With the ignition switch off, unplug the radar sensor and measure the resistance (ohms) using a multimeter between harness pins 4 and 7. The reading should be approximately 60 ohms. If it is not, the vehicle wiring should be investigated using procedures described by the manufacturer.

2.5 TROUBLESHOOTING WIRING HARNESSES

All wire harness connectors must be properly seated to maintain environmental seals. Push the mating connector until it seals with a click. When replacing an FLR-25 radar sensor, check that the wire harness connector is free of corrosion before plugging it into a new radar sensor. Check for corroded or damaged wiring connector problems such as opens or shorts to voltage or ground.

If the connector terminals are corroded, this may be an indication of water intrusion into the wiring system and possibly into the radar sensor. Replacement of the entire harness is recommended. If the terminals of the radar sensor are corroded, replacement of the radar sensor is recommended.

2.6 RESETTING THE RADAR SERVICE ALIGNMENT

A service alignment is recommended in the following situations:

- When an alignment fault is present;
- When a service radar is installed; and/or
- When the vehicle front end has been damaged.

Connect the FLR-25 radar to ACom Diagnostic Software to initiate the service alignment process. Bendix ACom Diagnostic Software is available on B2Bendix.com and is the only diagnostic tool that supports the service alignment process. The Configuration section in ACom Diagnostic Software has the option to start the *Radar Service Alignment*.

After initiating the service alignment via ACom Diagnostic Software, drive the vehicle (above 30 mph [48 kph]) until ACom Diagnostic Software displays the test results as *Alignment Successful*. The typical driven alignment time frame is 15 minutes; however, based upon conditions, it may take up to 60 minutes of driving time.

Alignment is optimal if you satisfy the following conditions during the drive:

- Keep the vehicle speed above 30 mph (48 kph).
- Keep a steady speed with minimal longitudinal or lateral accelerations during the drive.
- Plan routing with many stationary objects in the lanes next to the vehicle. For example, drive on a road where there are guardrails, light poles, or a turning lane with stopped vehicles. The alignment process may go faster if there are stationary objects on both right and left lanes next to the vehicle.

After ACom Diagnostic Software shows successful alignment, verify that the radar is properly aligned by checking that there are no alignment faults active in ACom. If this fault code is active, make sure the radar is mounted correctly and re-run the procedure. Note that any power cycle or interruption will cause the calibration to fail.

Refer to *BW5266, Bendix® Fusion™ 3: Radar (FLR-25™) and Camera (FLC-25™) Calibration and Alignment*, on B2Bendix.com for additional information.

2.7 FOLLOWING DISTANCE ADJUSTMENT SWITCH (OPTIONAL)

If the vehicle is equipped with the following distance adjustment switch and the following distance does not change after an adjustment is made, the switch, wiring, or a controller on the vehicle should be checked using the diagnostic procedures described by the vehicle manufacturer. The radar sensor receives the driver's desired following distance on the J1939 data communication link from a controller on the vehicle. No DTC will be set if the vehicle is not equipped with a following distance adjustment switch. *Refer to Table 5 for further information on configuration settings.*

2.8 CONFIGURING FUSION SYSTEM FOLLOWING DISTANCE ALERTS (FDA)

Changing the FDA configuration allows properly trained personnel to adjust both the FDAs and the following distance behind a detected forward vehicle using ACom Diagnostic Software. *Refer to Table 5 for all configuration specifications and see Figure 3 for an illustrated example.*

NOTE: Following Distance Mode 2 is the default setting. Following Distance Modes 0, 1, 3, and 4 may or may not be available depending on OEM. Refer to the applicable operator's manual on B2Bendix.com for assistance.

FDA Config. No.	Vehicle Speed Range	Fast Audible Alert (sec.)	Medium Speed Audible Alert (sec.)	Slow Audible Alert (sec.)	Following Distance (Mode 4) (sec.)	Following Distance (Mode 3) (sec.)	Following Distance (Mode 2) Default (sec.)	Following Distance (Mode 1) (sec.)	Following Distance (Mode 0) (sec.)
1	0 - 37 MPH	0.5	–	–	1.8	2.3	2.8	3.5	3.7
	38 - 52 MPH	0.5	0.875	1.125					
	> 52 MPH	0.5	1.0	1.5					
2	0 - 37 MPH	0.5	–	–	2.3	2.8	3.5	3.5	3.7
	38 - 52 MPH	0.5	0.875	1.125					
	> 52 MPH	0.5	1.0	1.5					
3	0 - 37 MPH	1.0	1.5	–	2.3	2.3	2.8	3.5	3.7
	38 - 52 MPH	1.0	1.5	2.0					
	> 52 MPH	1.0	1.5	2.0					
4	0 - 37 MPH	1.0	1.5	–	2.3	2.8	3.5	3.5	3.7
	38 - 52 MPH	1.0	1.5	2.0					
	> 52 MPH	1.0	1.5	2.0					
5	0 - 37 MPH	1.0	1.5	–	3.5	3.5	3.5	3.5	3.7
	38 - 52 MPH	1.0	2	3.0					
	> 52 MPH	1.0	2	3.0					
6	0 - 37 MPH	1.0	1.5	3.0	3.5	3.5	3.5	3.5	3.7
	38 - 52 MPH	1.0	2	3.0					
	> 52 MPH	1.0	2	3.0					
7	0 - 37 MPH	0.2	–	–	1.8	2.3	2.8	3.5	3.7
	38 - 52 MPH	0.2	–	–					
	> 52 MPH	0.2	–	–					
8	0 - 37 MPH	0.5	–	–	1.8	1.8	1.8	2.3	2.5
	38 - 52 MPH	0.5	0.875	1.125					
	> 52 MPH	0.5	1.0	1.5					
9	0 - 37 MPH	0.5	–	–	1.8	1.8	2.3	2.8	3.0
	38 - 52 MPH	0.5	0.875	1.125					
	> 52 MPH	0.5	1.0	1.5					

Table 5 – Configuring FDAs

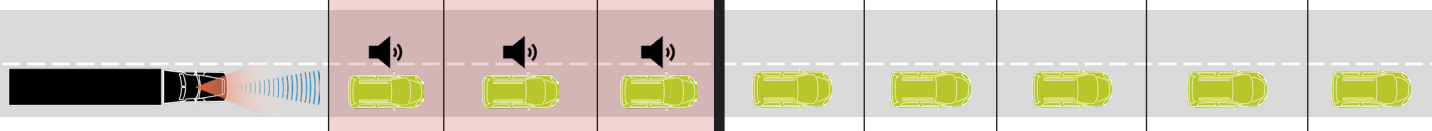
FDA Config. No.	Vehicle Speed Range	Fast Audible Alert (sec.)	Medium Speed Audible Alert (sec.)	Slow Audible Alert (sec.)	Following Distance (Mode 4) (sec.)	Following Distance (Mode 3) (sec.)	Following Distance (Mode 2) Default (sec.)	Following Distance (Mode 1) (sec.)	Following Distance (Mode 0) (sec.)
1	38 - 52 MPH	0.5	0.875	1.125	1.8	2.3	2.8	3.5	3.7
									

Figure 3 – FDA Config. No. 1 Example

For FDA Config. No. 1, if your vehicle is traveling between 38-52 mph, there will be a fast audible alert if your vehicle is within .5 seconds of the detected forward vehicle; there will be a medium speed audible alert if your vehicle is within .875 seconds of the detected forward vehicle; and there will be a slow audible alert if your vehicle is within 1.125 seconds of the detected forward vehicle. The faster the audible alert, the closer your vehicle is to the detected forward vehicle.

Each configuration may have up to five following distance modes (4, 3, 2, 1, and 0). Refer to the applicable operator's manual on B2Bendix.com for assistance in determining availability of following distance modes. Each mode adapts your vehicle speed to the speed of the detected forward vehicle to keep your vehicles the specified number of seconds apart from each other. For FDA Config. No. 1, Following Distance Mode 4 adapts your vehicle's speed to the speed of the detected forward vehicle to maintain a distance of 1.8 seconds between the two.

APPENDIX A – RADAR MOUNTING AND INSTALLATION

MOUNTING THE FLR 25 RADAR SENSOR



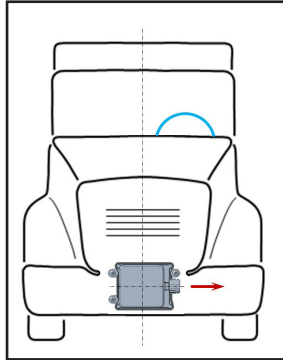
WARNING

Improper use of the Fusion system include repositioning from the original production line installation. Under no circumstances should the radar be removed or repositioned. The assembly shall always be mounted in the original OEM location. Failure to do so may result in a collision causing property damage, serious injuries, and/or death. If this location is not in the center of the vehicle, the mounting offset will need to be programmed through ACom Diagnostic Software.



CAUTION

Vehicle equipment, including bumpers, deer guards, etc., must not infringe upon the zone used by the radar sensor to emit and receive radar waves. Refer to Appendix A.3. Failure to comply with this requirement will impair the function of the radar. Only Bendix FLR-25 or vehicle OEM-approved covers and/or cover panels may be installed in front of the radar.



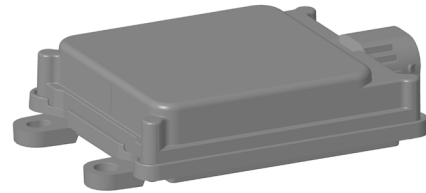
The radar sensor assembly is mounted to the front of the vehicle using a non-adjustable bracket. When mounting a radar sensor, the wire harness connector must always point towards the driver's side of the vehicle.

A.1 VEHICLE APPLICATIONS

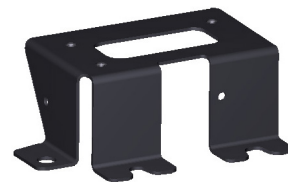
The radar sensor can be mounted and installed only on vehicles that have the Fusion system already installed.

A.2 REPLACEMENT PARTS

Replacement parts exist for the components shown below. Parts are available from any Bendix authorized parts supplier.



Radar Sensor

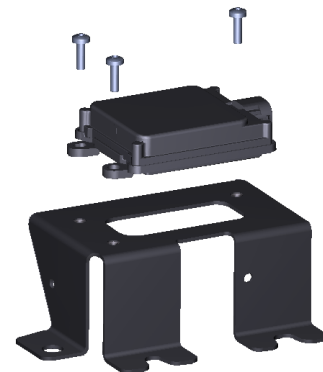


Radar Bracket

Provide the bracket part number (see label/stamp) when ordering replacements. Bracket will include three (3) mounting screws and the clinch nuts.

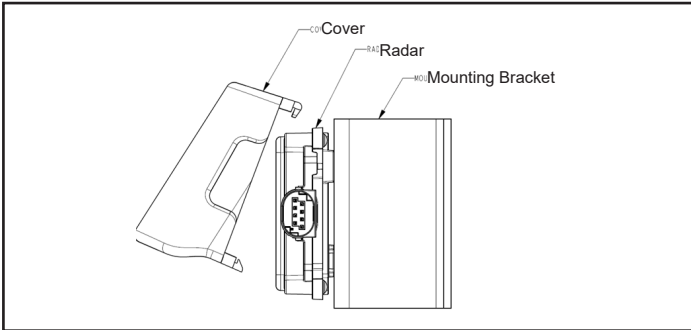
A.2.1 RADAR ASSEMBLY

This image shows the proper assembly sequence for the radar bracket assembly. For proper mounting, the screws must be torqued to 53 + 9 in-lb (6 + 1 N•m). Once assembled, the radar assembly is to be mounted to a defined location at the front of the vehicle with the FLR-25 radar sensor or OEM-approved cover in place.

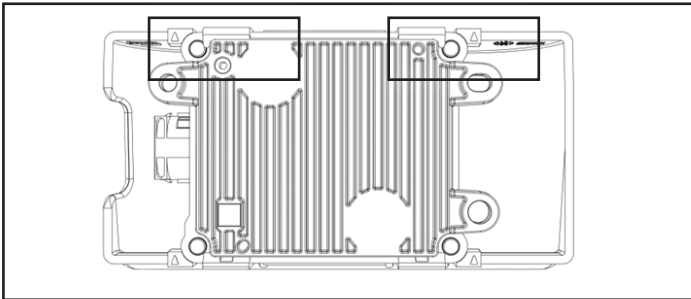


A.2.2 RADAR COVER INSTALLATION

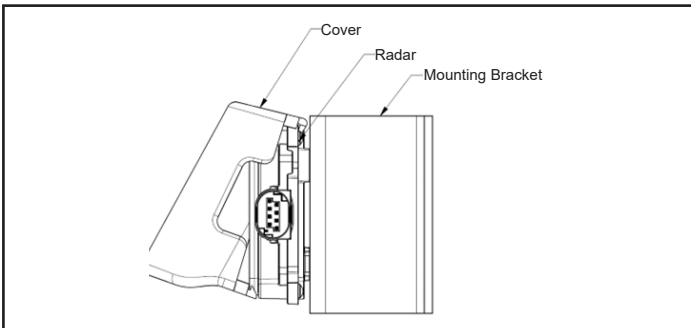
1. Position the cover near the radar.



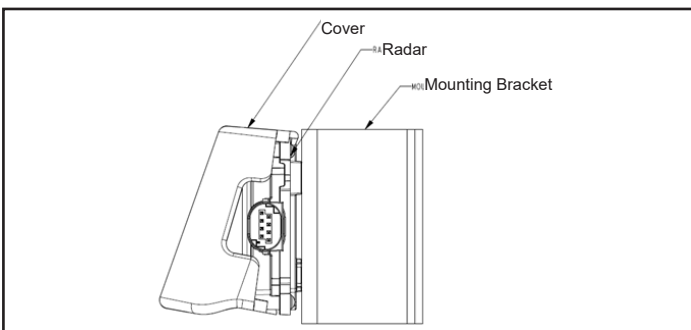
2. Insert both clips on top of the cover over the edge of the FLR-25 radar. The top clips index onto the FLR-25 radar.



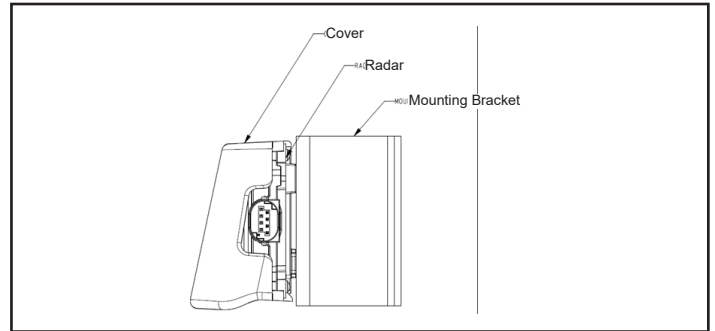
3. While keeping the top clips in place, rotate the cover downward to engage the lower clips with the edge of the radar.



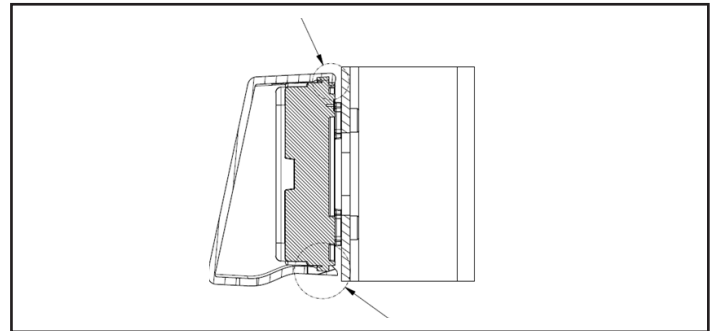
4. Keep pushing the clips to expand the bottom clips over the radar.



5. Keep pushing the clips until the lower clips snap over the metal base plate of the radar.

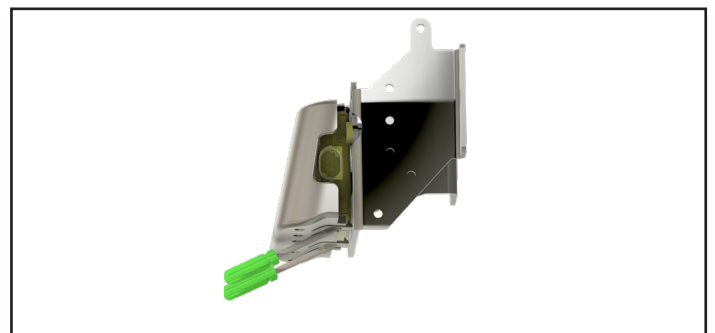


6. Ensure the cover is secure by manually checking for lack of play.

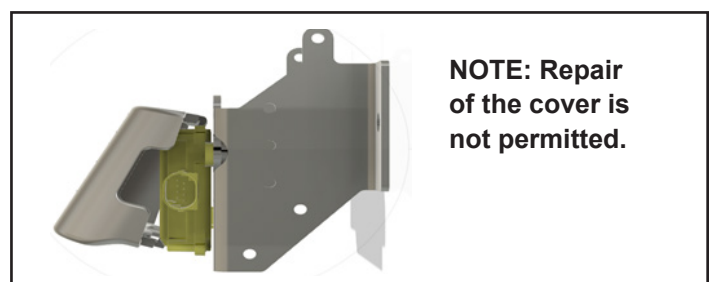


A.2.3 RADAR COVER REMOVAL

1. Insert a small flat-head screwdriver into a slot on the bottom of the cover near the lower clips.
2. Pry the clip downward releasing the clip over the edge of the radar base. This can either be done at the same time or one clip at a time.



3. Roll the cover up over the radar by repeating the steps of the installation in reverse order.



A.3 RADAR SENSOR MOUNTING CLEARANCE



Vehicle equipment, including bumpers, deer guards, etc. should not infringe upon the zone used by the radar sensor *shown in Figure 1* to emit and receive radar waves. Failure to comply with this requirement may impair the function of the collision mitigation system. Bendix recommends to only use OEM-approved and supplied radar covers with our radar products. OEM-supplied radar covers have been tested and approved for radar operation with minimum degradation or attenuation of the radar signal. Non-approved radar covers may lead to degradation of the driver assistance system. To qualify a new radar cover design, please contact the vehicle OEM.

NOTE: Bendix does not certify nor offer warranty on Fusion systems where system performance is affected by obstructions of any kind. Furthermore, Bendix will not certify or approve of any third-party bumper guards mounted on the front of the vehicle. Bendix does NOT recommend the removal of an Advanced Driver Assistance System (ADAS) from a vehicle. This document gives general guidelines that will work for most vehicles; exceptions may exist.

For proper operation of the FLR-25 radar sensor, adhere to the following guidelines:

- The radar sensor assembly should be OEM-installed on the vehicle following all OEM specifications.
- The radar's field of view must NOT have interference from any other vehicle components such as bumpers, cow-catcher bumpers, engine blankets, seasonal decorations, or any other commonly mounted front-of-vehicle components. The radar signal is emitted from the front of the sensor with a spreading beam. The approved OEM-provided radar cover is the only item that is to be placed in front of the radar. In order to ensure that no adverse interference is experienced from bumpers or other nearby vehicle equipment, a suitable clearance must be maintained around the radar. This clearance must be maintained regardless if the vehicle is stationary or in motion. *See the diagram below for a general guide and an example of how to calculate the zone required.* The front of the radar may not be accessible to measure from; it may be behind the OEM radar cover.



Forward deer/moose guards: *The field of view of both the radar and the camera need to be kept clear of any obstruction or the Fusion system performance may be affected. Bendix does not recommend the use of any forward vehicle guards that may interfere with a sensor's field of view, and use of such guards may void any Bendix warranty. Bendix includes the radar field of view diagram in the Service Data Sheet as a reference to help assist those fleets who are considering placement of such forward vehicle guards. Also, do not move the radar as it is designed by Bendix and the OEM to operate in the location placed by the OEM during factory installation.*

KEEP-OUT ZONE

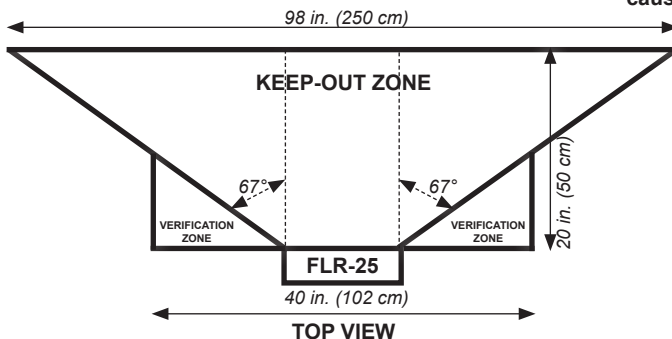
Length: 98 in. (250 cm)

Height: 15 in. (38 cm)

Width (distance from radar midpoint): 20 in. (50 cm)

This area needs to be kept free of any materials or objects that may disturb the radar function. For example, metallic parts shall not violate this Keep-out Zone.

Azimuth: 67° | Elevation: 15°



NOTE: Drawings are representative only and not to scale. Dimensions are accurate.

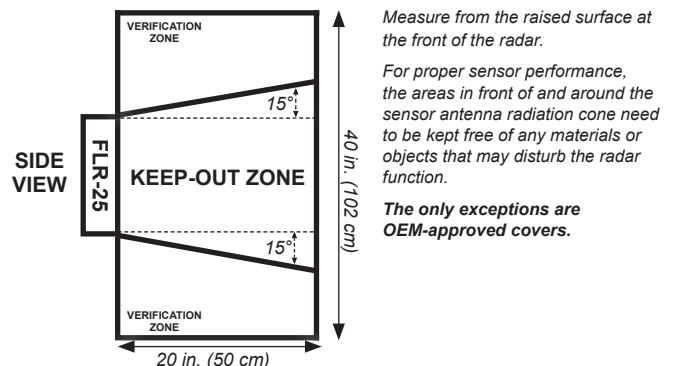
VERIFICATION ZONES

Length: 40 in. (102 cm)

Height (on both the top and bottom of the Keep-out Zone): 12.5 in. (31.75 cm)

Width (distance from radar midpoint): 20 in. (50 cm)

Any objects present in these areas (on both the top and bottom of the Keep-out Zone) should be verified by the manufacturer of the intruding object not to cause degradation of system performance. Failure to do so can result in impaired functions of the system and can result in false-positive activation of the system or a collision causing property damage, serious injuries, and/or death.



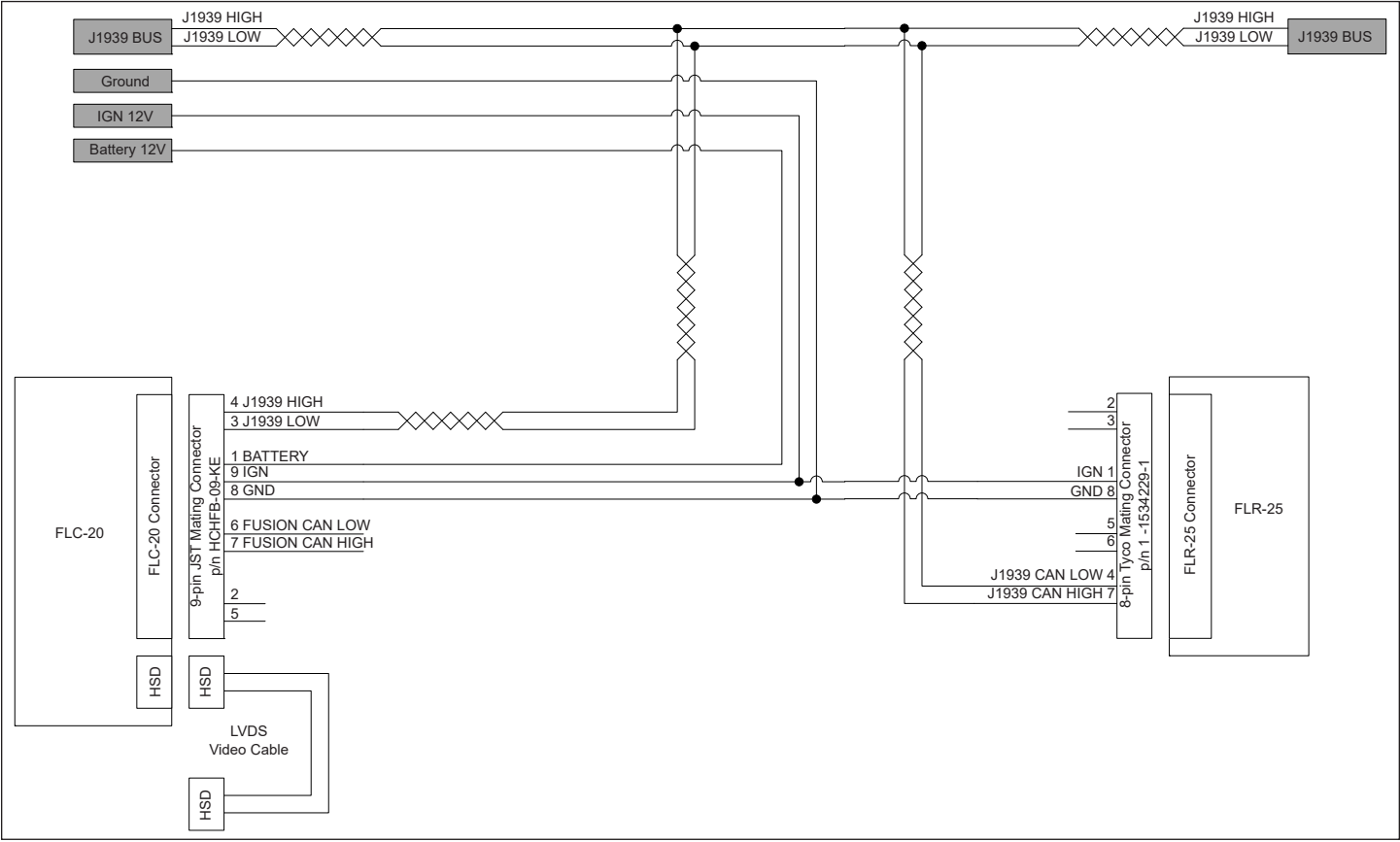
Measure from the raised surface at the front of the radar.

For proper sensor performance, the areas in front of and around the sensor antenna radiation cone need to be kept free of any materials or objects that may disturb the radar function.

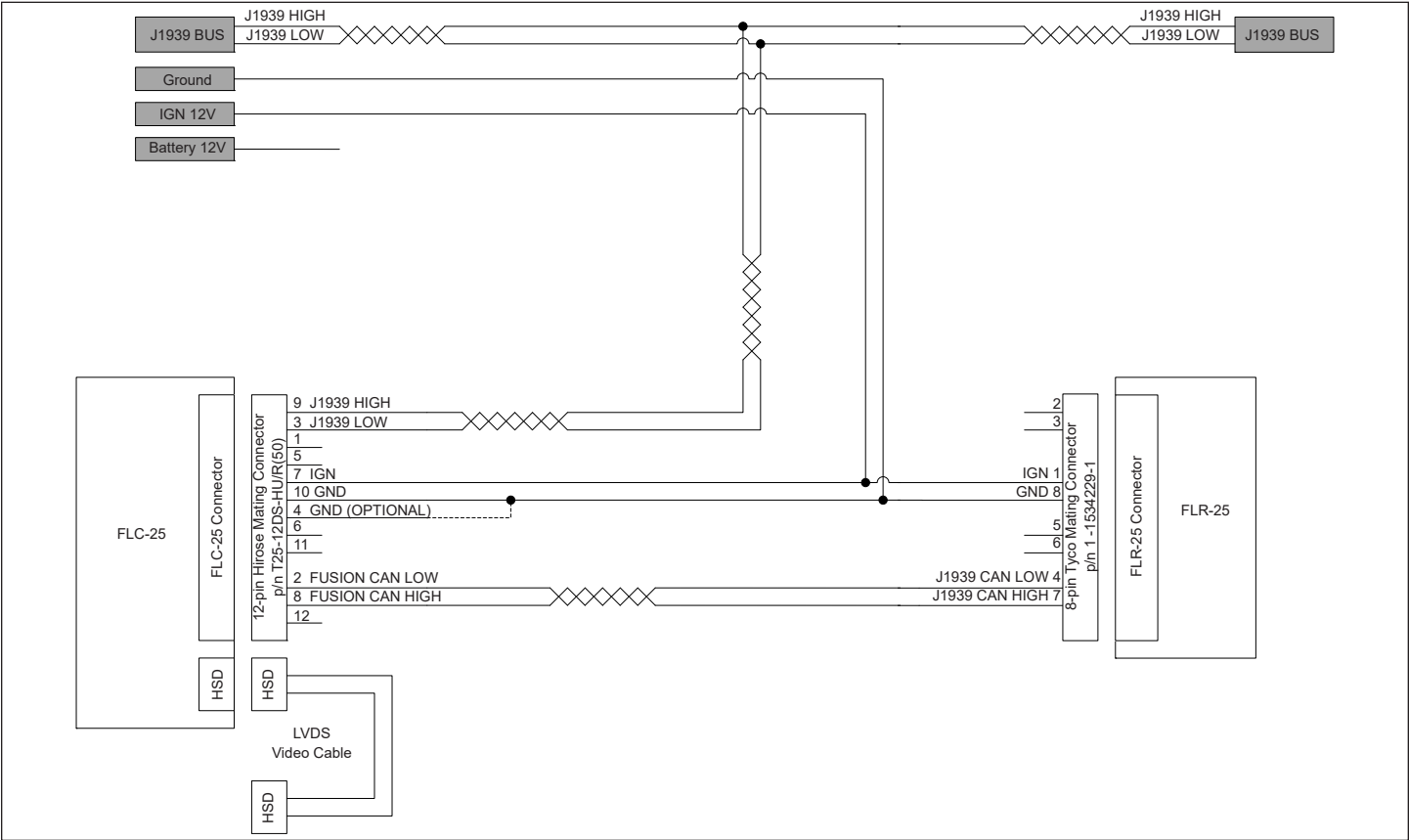
The only exceptions are OEM-approved covers.

APPENDIX B – SYSTEM WIRING SCHEMATIC

B.1 BENDIX FUSION 2.10: FLR-25 AND FLC-20 WIRING SCHEMATIC



B.2 BENDIX FUSION 3.0: FLR-25 AND FLC-25 WIRING SCHEMATIC



For the latest information, and for downloads of ACom Diagnostic Software, visit *B2Bendix.com*. For full DTC information, refer to *B2Bendix.com > Services & Support > Diagnostic Trouble Code Search*.

NOTES

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

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